Applying the PAY Rule™ with the Historical Data

Thus far, we have looked at applying a 4 percent initial withdrawal rate to the different retirement spending strategies. In such cases, we did not use a PAY Rule to calibrate the level of downside risk, as the initial spending rate was always the same.

Now, we are ready to consider the impact of controlling downside wealth outcomes and allowing the initial withdrawal rate to vary to maintain the same downside risk across strategies. Exhibit 6.20 provides results for the various spending strategies using a PAY Rule that allows for a 10 percent chance that real wealth has fallen below $15,000 from an initial $100,000 by year thirty of retirement.

This analysis is again conducted using a 50/50 portfolio of stocks and bonds and is based on rolling periods from history. For constant inflation-adjusted spending, the supported initial spending rate is 4.18 percent. To be clear, this is higher than the SAFEMAX value of 4.03 percent. This is because the PAY Rule is not that spending falls to zero with 100 percent probability by year thirty of retirement, but that there is a 10 percent chance that wealth falls below $15,000 by year thirty. It is a different criterion. Spending can be higher because we allow for a 10 percent chance that real wealth will fall below a threshold that is not too high to offset the higher accepted downside probability risk.

For constant inflation-adjusted spending, spending stays at this initial level in real terms throughout retirement across the distribution. Remaining wealth after thirty years ranges from just below the $15,000 limit allowed in year thirty at the tenth percentile to as much as $310,000, or three times more than the initial wealth balance, in the ninetieth percentile. In the median outcome, real wealth has grown by about 10 percent.

For this baseline strategy, spending does not increase or decrease in any of the scenarios, as wealth has not been depleted for the circumstances shown in the exhibit. Therefore, changes in real spending, changes in spending relative to the baseline, and downside spending volatility are all zero.